

3.2 Equilibrium of a rigid body

Further Mechanics · 2026 examination · Concept guide and original exam-style practice

Understand the concept

Rigid-body equilibrium requires both resultant force zero and resultant moment zero. Weight acts through centre of mass; contact reactions and friction depend on geometry. Take moments about a point that removes unknown forces to solve efficiently, then check force components and physical sign.

Worked example: follow the reasoning

A 10 N weight 0.8 m from pivot is balanced by force F at 0.4 m on opposite side: $F=20$ N.

Structured practice

Answer in the space provided. Show working and units where relevant.

- 1 A horizontal uniform beam 4 m long has weight 80 N at centre; a support at one end and upward force at other hold it. Find upward force at far end. [3]

Final answer: _____

Answers and explanation

Question 1 [3 marks]

Take moments about support: $F \times 4 = 80 \times 2$ [2]; $F = 40$ N [1].

Self-assessment

Compare each credited step with your own answer. Fix the first reasoning or calculation error; then retry without looking at the solution.

Syllabus: Further Mathematics 9231, 2026–2027.

<https://www.cambridgeinternational.org/Images/697357-2026-2027-syllabus.pdf>

Original FagusED teaching material. Question layout and mark presentation informed by the May/June 2026 papers supplied by the user. The practice question is newly written and is not a Cambridge past-paper question.